

pertaining only to that programme to the user.

Use of Wi-Fi modem enables multiple users to use the IPTV. Introduction of fibre-optic broadband connection enhances the data rate significantly and facilitates multiple users to access programmes of their choice simultaneously.

Challenges in TV receiver design

Barring IPTV, all other standards require RF systems at the input end of a TV receiver. All digital TV base-band signals utilise a digital bit-stream of more or less similar structure, except for DVB-T/T2, which requires additional OFDM step. The difference lies in the use of certain RF carrier frequencies and code variations specific to the standard. The software for signal extraction would therefore differ in case of DVB-S/S2, DVB-C/C2 and DVB-T/T2.

This is not a big problem, as the programme can be altered by input data itself as per requirement. As of today, DVB-C/C2, T/T2 and S/S2, all need a separate and unique STB, differing from each other mostly in RF circuitry, to interface to the user TV. IPTV too needs a different STB, as it does not depend on the RF carrier.

Digital cable and satellite channels, made available by private service providers, utilise encryption to facilitate collection of subscription from users. DVB-T/T2 or DTT, on the other hand, is an FTA broadcasting system radiating on standard UHF carriers. A TV with built-in DVB-T/T2 tuner can therefore be connected to UHF Yagi antenna to receive signals, without the need of a separate DVB-T/T2 STB.

In principle, RF sections of the STB for DVB-C/C2, DVB-T/T2 and DVB-S/S2 standards are similar, in that all of these have to be able to recover digital bit-stream from a VHF/UHF carrier. Separation and reconstruction of video and audio signals can be accomplished by a unified decoder under program control. This paves the way for the design of a unified digital TV receiver called integrated digital TV (iDTV) receiver.

Paid channels managed by service providers can be taken care of by incorporation of user-replaceable SIM

(subscriber identity module) cards, as used in cellphones. Addition of adequate processing power has resulted in the smart TV, which is capable of handling signals available from a USB port, Wi-Fi or Internet port, in addition to satellite, terrestrial and cable.

The smartphone as a user terminal

Arrival of the cellphone in the year 2000 and its amalgamation with the palm-top computer has made it emerge as the smartphone, and has revolutionised the concept of information interchange. The smartphone is capable of handling audio as well as video information, and sharing the same between users.

Internet connectivity has ushered in an era where a large amount of all sorts of information is available at one's fingertips. Therefore any broadcast technology has to recognise the dominance of the smartphone in public life and ensure compatibility, for wider acceptance.

One mode of watching DVB-T2 programmes would be via a dongle containing all RF circuits pertaining to UHF TV, including bit-stream handler and decoder. The dongle requires a UHF antenna to pick up radiated signals, and an Android app needs to be downloaded to the device for the DTT dongle to work. Many Wi-Fi routers are available in the market, which can be simply plugged to a power point at home to receive DTT signals, to serve Android devices, located nearby.

Conclusion

Today, public demand is towards video information and entertainment, and preferred choice of the user terminal is the ever-present smartphone. Any technology, to get a hold on public attention, needs to be compatible to the cellphone. The DTH satellite service is quite popular, while the simple DVB-T2 system that uses the standard Yagi antenna for reception has not penetrated the market due to lack of information as well as the relative absence of DTT receivers.

Cable TV competes with satellite TV, and blackout of Ka and Ku band signals during rains gives an edge to cable TV. IPTV and cable TV would be at par in this respect. **EFY**



POCKET SIZE DIGITAL MULTIMETERS

108B+ TRMS



101B+



Pocket Size

- 3½ Digits 4000 Counts (101B+)
- 3½ Digits 6000 Counts (108B+ TRMS)
- Auto Ranging
- TRMS Measurement (108B+ TRMS)
- LCD with Backlight
- Upto 1000V DC
- Upto 750V AC
- Upto 10A DC / AC
- Upto 40MΩ (101B+)
- Upto 60MΩ (108B+ TRMS)
- 4.000nF ~ 4.000mF (101B+)
- 9.999nF ~ 9.999mF (108B+ TRMS)
- 99.99Hz ~ 20.00MHz
- 0.1% ~ 99.9% Duty Cycle
- RELΔ (101B+)
- -20°C ~ 1000°C (-4°F ~ 1832°F) (108B+ TRMS)
- K Type Thermocouple (Upto 260°C) (108B+ TRMS)
- Diode Test, Audible Continuity, Data Hold, Auto Power Off, & Low Battery Indicator

Meco Meters Private Limited
 Plot No. EL-60, MIDC Electronic Zone,
 TTC Industrial Area, Mahape,
 Navi Mumbai - 400710 (INDIA)
 Tel : 0091-22-2767 3300 (Board), 2767 3311-16 (Sales)
 Fax : 0091-22-27673310
 Email : sales@mecoind.com Web : www.mecoind.com
 Connect With Us :